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ORBIT NORTH

Space Mission Design and Operations — 12-week format — 3 credits

ORBIT NORTH
Winter 2027 (draft)

Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

GNG 3100 · SYS 5186 · ORBIT NORTH · Issue 1.0 · 2026-11-08

Classification: Internal

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Space Mission Design and Operations — 12-week format — 3 credits GNG 3100 — Topics in Engineering I · SYS 5186 — Advanced Topics in Systems Engineering · Winter 2027 (draft)

> **Draft.** This is the 12-week version of the ORBIT NORTH summer intensive: the same outcomes, material and assessments, paced across a regular term. Rooms, times and dates are to be confirmed.

Course Information

Format. 3-credit course over a **12-week term**, delivered in person at the University of Ottawa. Each regular week has two sessions:

Session	Length	What happens	Location
A — Lecture and seminar	1.5 h	Short lecture on the week's topic, discussion, guest speakers	Classroom, TBC
B — Studio	3 h	Team work in the Concurrent Design Facility (Weeks 1–9) or the mission control room (Weeks 10–12)	SpaceCDF lab · uOttawa Cyberrange

The two **simulation weeks (Weeks 11 and 12)** replace the studio with one **extended 6-hour block** at the Cyberrange, because a simulated operations shift cannot be split. A final 3-hour session for the debrief and final mission review is held in the last week of classes or early in the examination period. A one-week **reading break** falls after Week 6.

Mode. In person. The CDF and simulation components require physical attendance; recordings of the short lectures, the course videos and the open textbook are available online.

Instructor Information

Lead Instructor. Dr Franz Newland · `fnewland@uottawa.ca` · SEDTI, Faculty of Engineering, University of Ottawa.

Industry engagement. Guest speakers from Canadian space organisations. Confirmed speakers will be posted in Brightspace.

Communication. In person before and after class, or by email. Please allow 2 business days for replies. Please read this syllabus in full before emailing a question — many answers are here, and students may be redirected to the syllabus when the answer is already available.

Intended Audience

This course supports differentiated outcomes for undergraduate (GNG 3100) and graduate (SYS 5186) students. It is designed to be interdisciplinary; students from any technical background are welcome and will be deliberately mixed.

Official Course Description

ORBIT NORTH is a space systems course delivered by the University of Ottawa School of Engineering Design and Teaching Innovation (SEDTI). Students move from the Canadian space ecosystem and its regulatory

environment, through a complete preliminary CubeSat mission design produced in a Concurrent Design Facility, to the operator seat in a live mission simulation. The course is designed as a primer for professional practice in the Canadian and international space sector for students with no prior space background.

Additional Course Description

The term follows the same three-part arc as the summer intensive.

Part 1 — Context and concept (Weeks 1–4). The Canadian space ecosystem, the concurrent design method and the SpaceCDF platform; from a mission need to concept v0 and a first concept of operations, presented at an informal Mission Concept Review (MCR) at the end of Week 2. Week 3 closes the MCR actions and locks the concept; Week 4 covers the Canadian regulatory landscape in depth — the Remote Sensing Space Systems Act, spectrum licensing through ISED, export controls, debris mitigation and registration — and produces the team's regulatory brief.

Part 2 — Concurrent design (Weeks 5–9). One CDF week per design stage: System-V and the requirements baseline (Week 5; PRR and SRR are explicitly collapsed into this week); orbit, power, AOCS and thermal (Week 6); comms, structure and propulsion with two formal trade studies (Week 7); and integration, V&V, cost and risk closure (Week 8). Week 9 is the PDR-style review (Preliminary Design Review). The weekly rhythm leaves time between CDF sessions for each discipline lead to do the analysis that the intensive compresses into evenings.

The systems-engineering frameworks themselves — System-V, requirements engineering, functional decomposition, margin policy, trade studies, risk, and planning — are learned **inside** the CDF, because the SpaceCDF platform embeds the SE process scaffolding, the risk register, the planning baseline, and a regulatory-check module as live tools.

Part 3 — Operations (Weeks 9–12). The ground segment and the mission control system (Week 9), operations training and dry runs at the uOttawa Cyberrange (Week 10), and two simulation weeks flying EOSAT-1: LEOP and commissioning (Week 11), then nominal operations and contingency (Week 12). In each simulation one team operates while another observes and documents; the teams then swap. The course closes with a structured debrief and a cohort-internal final mission review.

There are no required textbooks to purchase. All readings, tools, and software access — including SpaceCDF and the open textbook *Concurrent Spacecraft Design* — are provided through Brightspace and the uOttawa Cyberrange environment.

Indigenous Affirmation

ANISHINÀBE. Ni manàdjiyànànig Màmìwininì Anishinàbeg, ogog kà nàgadawàbandadjig iyo akì eko weshkad. Ako nongom ega wìkàd kì mìgiwewàdj. Ni manàdjiyànànig kakina Anishinàbeg ondaje kaye ogog kakina eniyagizidjig enigokamigàg Kanadàng eji ondàpinangig endàwàdjin Odàwàng. Ninisidawinawànànig kenawendamòdjig kije kikenindamàwin; weshkinìgidjig kaye kejeyàdizidjig. Nigijeweninmànànig ogog kà nìgànì sòngideyedjig; weshkad, nongom; kaye àyànikàdj.

ENGLISH. We pay respect to the Algonquin people, who are the traditional guardians of this land. We acknowledge their longstanding relationship with this territory, which remains unceded. We pay respect to all Indigenous people in this region, from all nations across Canada, who call Ottawa home. We acknowledge the traditional knowledge keepers, both young and old. And we honour their courageous leaders: past, present, and future.

Inclusion

The University of Ottawa aims to be an equitable and inclusive institution, actively participating in ensuring the well-being of students, personnel, and faculty members. The University is committed to eliminating obstacles to student inclusion in accordance with the Ontario Human Rights Code. If you would like an alternative name or pronouns used for you during the course, please email the instructor. If you have experienced discrimination or harassment, you can seek confidential assistance through the University Human Rights Office.

The course combines long team studios with a live simulation in which the clock keeps running. If health, neurodiversity, caregiving, work or financial considerations affect your ability to take part fully, please reach out in the first two weeks of term so we can plan together.

Course Learning Outcomes

Learning outcomes are differentiated between the undergraduate (GNG 3100) and graduate (SYS 5186) codes. All students pursue the shared outcomes; graduate students additionally demonstrate the graduate-level outcomes across the same activities.

Shared outcomes — both codes

- **LO 1.** Describe the Canadian space ecosystem (past, present, emerging) and situate a given mission concept within it.
- **LO 2.** Apply the systems-engineering scaffolding embedded in a Concurrent Design Facility to structure a space mission concept through scope, ConOps, functional architecture, and preliminary budgets, drawing on ECSS and the NASA Systems Engineering Handbook.
- **LO 3.** Navigate Canadian satellite regulatory obligations — the Remote Sensing Space Systems Act, radiocommunication licensing, export controls, and debris mitigation — at a level sufficient to identify when professional advice is required.
- **LO 4.** Contribute substantively to a Concurrent Design Facility intensive design activity, taking a discipline role and delivering a complete preliminary CubeSat mission design across all principal subsystems.
- **LO 5.** Maintain a risk register and a planning baseline inside the CDF tool through the full design week.
- **LO 6.** Execute spacecraft operations procedures in a mission control environment, including nominal operations, planning, and anomaly response.
- **LO 7.** Collaborate in interdisciplinary student teams; give and receive structured technical critique.
- **LO 8.** Reflect critically on your own learning trajectory and on the ethical dimensions of space activity.

Additional graduate outcomes — SYS 5186 only

- **LO G1.** Critically evaluate architecture trade-offs using quantitative methods (margin-based budgets, risk-adjusted comparison) and defend the evaluation in writing and in a review setting.
- **LO G2.** Lead a sub-area of the team's mission design and operations, coordinating contributions from other team members and managing interface definition.
- **LO G3.** Synthesise lessons learned from the simulation into a structured improvement proposal grounded in the systems engineering and human-factors literature.
- **LO G4.** Situate your experience in the course within at least one strand of the scholarly literature on space systems engineering, operations, or new-space practice.

How these outcomes connect to the Master of Systems Engineering

For SYS 5186 students, ORBIT NORTH counts towards the program learning outcomes (PLOs) of the Master of Systems Engineering; GNG 3100 work is marked with the same rubrics. Each course outcome below builds one or more of the program's course-level outcomes (CLOs), and each CLO belongs to a PLO. The same five-level rubric scale is used throughout the program: 1 Initial thoughts, 2 First engagement, 3 Deeper engagement, 4 First leadership, 5 Deeper leadership.

Course outcome	Program CLOs	PLOs
LO 1	7.4 Ecosystem navigation, 6.1 Domain learning agility	PLO6-MEng Applied project, PLO5 Interdisciplinary & digital (MBSE)
LO 2	1.1 Problem framing & need analysis, 1.2 Requirements & architecture, 3.1 Margin & budget management, 5.1 Standards: select & tailor	PLO1 SE principles, PLO2 Design, V&V, PLO4 Professional practice
LO 3	5.4 Controlled / cleared practice, 4.3 Ethics & governance	PLO4 Professional practice
LO 4	2.4 Concurrent design participation, 3.2 Trade study execution, 4.2 Multidisciplinary collaboration, 7.1 Cross-domain translation	PLO5 Interdisciplinary & digital (MBSE), PLO1 SE principles, PLO4 Professional practice, PLO3 Communication
LO 5	3.4 Risk management, 3.1 Margin & budget management	PLO4 Professional practice, PLO1 SE principles
LO 6	1.4 Decision under uncertainty, 2.3 Digital twin & simulation, 6.1 Domain learning agility	PLO1 SE principles, PLO5 Interdisciplinary & digital (MBSE)
LO 7	4.2 Multidisciplinary collaboration, 4.4 Communication & facilitation	PLO4 Professional practice, PLO3 Communication
LO 8	6.2 Reflective practice, 6.3 Portfolio curation, 4.3 Ethics & governance	GDLE6 Reflective practice thread (limits of knowledge), PLO4 Professional practice
LO G1	3.2 Trade study execution, 3.1 Margin & budget management, 4.4 Communication & facilitation	PLO1 SE principles, PLO3 Communication
LO G2	3.3 Technical decision leadership, 1.2 Requirements & architecture	PLO4 Professional practice, PLO2 Design, V&V
LO G3	6.2 Reflective practice, 7.1 Cross-domain translation	GDLE6 Reflective practice thread (limits of knowledge), PLO3 Communication
LO G4	6.1 Domain learning agility, 6.2 Reflective practice	PLO5 Interdisciplinary & digital (MBSE), GDLE6 Reflective practice thread (limits of knowledge)

Teaching Philosophy

My role on this course is to accompany you into a professional community. I bring standards, tools, and a network of colleagues — you bring judgement, curiosity, and the willingness to make decisions under uncertainty. The most important thing we will do together is practise thinking about missions as whole systems, and practise making decisions when the engineering is only part of what matters. You will be asked

to do things you don't yet feel qualified to do. That is not an accident; it is the core of the course. Many space projects involve doing something for the first time.

Teaching Strategies and Approaches

- Short, concentrated lectures (usually ≤ 45 minutes) anchored to open NASA / ECSS material and the open textbook, used to set up the week's studio. Short course videos with practice questions accompany each topic.
- A nine-week Concurrent Design Facility thread in which each student takes a discipline role and the team converges on a complete preliminary mission design, supported by the SpaceCDF platform's embedded SE process, risk, planning, and regulatory-check modules.
- Structured space-industry guest interactions throughout.
- A live, team-based mission simulation at the uOttawa Cyberrange, with alternating operator and observer roles.
- Peer-critique sessions structured around real review gates (Mission Concept Review, Preliminary Design Review).
- Individual reflective writing throughout the course.

A note on review-gate terminology

The course uses NASA terminology: the *Mission Concept Review (MCR)* is the end-of-Pre-Phase-A gate, and the *Preliminary Design Review (PDR)* is the end-of-Phase-B gate. ECSS reuses the abbreviation "MCR" for the Mission Close-out Review at the end of Phase E; we do not use that meaning here. The Phase A intermediate reviews (Preliminary Requirements Review, System Requirements Review) are explicitly **collapsed into CDF Week 1 (Week 5)** in this course; their content is delivered inside the SpaceCDF requirements module.

Required Materials

All required readings will be provided. Core open references students should bookmark:

- NASA *CubeSat 101 — Basic Concepts and Processes for First-Time CubeSat Developers* (`https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf`).
- NASA *Systems Engineering Handbook* (SP-2016-6105 Rev 2) (`https://www.nasa.gov/reference/systems-engineering-handbook/`).
- ECSS-E-ST-10C Rev.1 — *System Engineering General Requirements* (freely available from ECSS).
- Cal Poly CubeSat Design Specification, Rev 14 (CDS Rev 14).
- SpaceCDF platform — web access and Learner's Workbook provided at the start of the course.
- Optional self-study: the Space Systems Engineering curriculum (Lisa Guerra, NASA, Texas Space Grant) at `https://spacese.spacegrant.org` . The course no longer follows this module-by-module — its content is delivered through the SpaceCDF tool — but it remains a useful reference.

Assessment Strategy

Assessment is portfolio-based. The table below is the default weighting for GNG 3100. SYS 5186 weightings differ as noted. Detailed rubrics will be posted in Brightspace at the start of the course.

Assessment	Type	GNG 3100	SYS 5186	Timing
Pre-course readiness quiz	Individual	5 %	3 %	Week 1
Mission Concept Review (MCR, informal)	Team + Individual	15 %	15 %	Week 2
CDF daily worksheets and participation	Individual	10 %	5 %	Weeks 5 – 8
Preliminary Design Review (PDR-style review)	Team + Individual	20 %	22 %	Week 9
Operations training quiz	Individual	5 %	2 %	Week 10
Simulation performance — operator shift	Team	15 %	18 %	Weeks 11 – 12
Observer critique report	Team + Individual	10 %	10 %	Weeks 11 – 12
Final mission review	Team	10 %	5 %	Final session
Individual reflective portfolio (UG: structured reflection; Grad: critical literature-informed analysis, 2500 – 3000 words)	Individual	10 %	20 %	1 week after the final session
Total		**100 %**	**100 %**	

What the major assessments assess

The major assessments are marked with the program rubrics listed below (posted in Brightspace). The quizzes and the daily CDF worksheets are marked on their own keys.

Assessment	Course outcomes	Program rubrics (CLO → PLO)
Mission Concept Review (MCR, informal)	LO 1, LO 2, LO 7	1.1 Problem framing & need analysis → PLO1; 4.1 Stakeholder engagement → PLO3; 4.4 Communication & facilitation → PLO3
Preliminary Design Review (PDR-style)	LO 2, LO 3, LO 4, LO 5, LO G1, LO G2	1.2 Requirements & architecture → PLO2; 3.1 Margin & budget management → PLO1; 3.2 Trade study execution → PLO1; 3.4 Risk management → PLO4; 2.4 Concurrent design participation → PLO5; 4.4 Communication & facilitation → PLO3
Simulation performance — operator shift	LO 6, LO 7, LO G2	1.4 Decision under uncertainty → PLO1; 2.3 Digital twin & simulation → PLO5; 4.2 Multidisciplinary collaboration → PLO4; 3.3 Technical decision leadership → PLO4
Observer critique report	LO 6, LO 7, LO G3	4.4 Communication & facilitation → PLO3; 4.2 Multidisciplinary collaboration → PLO4
Final mission review	LO 4, LO 6, LO G2	1.3 Integration & verification → PLO2; 4.4 Communication & facilitation → PLO3
Individual reflective portfolio	LO 8, LO G3, LO G4	6.2 Reflective practice → GDLE6; 6.3 Portfolio curation → GDLE6; 4.3 Ethics & governance → PLO4

Level expected

- GNG 3100: Level 3 (Deeper engagement) is the target for an A.
- SYS 5186: Level 3 (Deeper engagement) is the baseline; an A shows a substantial share of Level 4 (First leadership).

Time Commitment

A 3-credit course carries the same total workload whether it runs as an intensive or across a term. Expect:

- 4.5 hours of scheduled contact in a regular week (1.5 h lecture and seminar, 3 h studio); 6 hours in each simulation week.
- About 5 – 6 hours of preparation and team work outside class each week: readings and videos before the lecture, discipline analysis between CDF studios, and procedure study before the simulations.
- Heavier weeks before the two reviews (Weeks 2 and 9) and before each simulation (Weeks 11 and 12). Plan for them.

Team work between studios matters: in the CDF weeks your discipline's numbers are an input to everyone else's.

Language Expectations

All course materials — slides, videos, textbook, workbook, procedures — are provided in English and French, and the SpaceCDF and operations tools can be used in either language. The language of instruction for each section is the one shown in the course timetable. You may write any work, give any presentation, and speak on the simulation voice loop in either official language. Communication effectiveness is assessed throughout the course; selecting the appropriate choice of language and register for a given audience is part of that assessment.

Late Assignments and Grade Revision

All assignments should be submitted by their posted due date and time. In the case of illness or other serious situations, contact the instructor as early as possible; university regulations require late submissions due to illness to be supported by a medical certificate. Formal grade revision requests must follow Academic Regulation A-9. Please contact me first for clarification before pursuing a formal appeal.

Participation Guidelines

You can expect your facilitators to:

- Treat you with respect and hear and value your viewpoints.
- Provide constructive feedback and criticism, returned within 1 week of submission.
- Deliver sessions as scheduled; communicate any adjustments through Brightspace and email.
- Respond to inquiries through appropriate channels within 1 business day during teaching weeks.

Learners are expected to take responsibility for their own success:

- Manage your time; the CDF weeks build on each other, so falling behind one week costs the whole team the next.
- Come to each session prepared, having completed required readings and any pre-work.

- Participate positively in team activities — you cannot pass this course as a lone operator.
- Demonstrate academic integrity; all generative-AI use must be declared using the AIG attribution framework introduced in Week 1.

Course Calendar

Weeks are indicative. Dates, rooms and any adjustments will be confirmed in Brightspace. Lecture numbers (L01–L14) refer to the course slide decks.

Part 1 — Context and concept

Week	Lecture and seminar (1.5 h)	Studio (3 h)	Deliverables
1	**L01 Canada in space and course onboarding.** Past, present and emerging Canadian space activity; course arc; teams, roles and tools.	**L02 The concurrent design method.** Regulatory teaser; SpaceCDF accounts and tour; teams open a mission file and draft a problem statement.	Readiness quiz; team formed; mission area chosen
2	**L03 From need to concept.** Problem statements, stakeholders, MoE / MoP / TPM, trade studies, ConOps.	Mission alternatives trade; ConOps v0; first parametric budgets; risk register opened. **Informal MCR** in the last hour.	Concept v0; ConOps v0; **informal MCR**
3	**L04 Closing the MCR and locking the concept.** Review actions, concept lock, top-level requirements, programmatic envelope.	MCR action closure; concept v1; top-level requirements; risk register v0.	**Concept v1 locked**
4	**L05 The Canadian regulatory landscape.** RSSSA; ISED CPC-2-6-02 and ITU coordination; Export Control List, Controlled Goods, ITAR / EAR; debris mitigation (IADC 25-year, FCC 5-year); UN Registration Convention.	Regulatory brief; draft ISED application; regulatory risks entered in the register.	Regulatory brief (team, 2 pp.); draft ISED application

Part 2 — Concurrent design

Week	Lecture and seminar (1.5 h)	Studio (3 h)	Deliverables
5	**L06 The System-V, requirements and first budgets.** Tailoring; lifecycle gate map; SMART requirements; functional decomposition; mass and power margins.	**CDF 1.** Requirements baseline; function tree; first mass, power, data and link budgets.	CDF worksheet 1; requirements baseline
6	**L07 Orbit, power, attitude and thermal.** Orbit trade; array and battery sizing; AOCS architecture; thermal balance.	**CDF 2.** Orbit selection; power, AOCS and thermal decisions logged.	CDF worksheet 2; subsystem decisions logged
—	*Reading break*		
7	**L08 Communications, structure, propulsion and the formal trades.** Link	**CDF 3.** Link budget and ground-station plan; regulatory check on	

Week	Lecture and seminar (1.5 h)	Studio (3 h)	Deliverables
	budget; band and licensing; structural margins; onboard data handling; propulsion.	the band; link-band and propulsion trade studies; equipment selection.	CDF worksheet 3; equipment list; two trade-study reports
8	**L09 Integration, verification, cost and risk.** N^2 interfaces; V&V matrix; environmental tests; cost estimation; FMECA and reliability.	**CDF 4.** Interface closure; V&V matrix; test plan; cost and bill of materials; risk closure; SpaceCDF optimiser; ECSS document export.	CDF worksheet 4; interface and V&V matrices; cost; ECSS document set
9	**L10 From design to operations.** EOSAT-1; mission phases; passes; PUS packets and services; a telecommand's life; the mission planner.	**PDR-style review (Preliminary Design Review)** with review-board questions and action capture.	**PDR delivered**

Part 3 — Operations · uOttawa Cyberrange

Week	Lecture and seminar (1.5 h)	Studio	Deliverables
10	**L11 The control room.** Positions, the voice loop, anomaly response, FDIR severity and escalation, shift handover.	**Operations training (3 h).** MCS familiarisation by position; pass planning; voice-loop drills; dry runs of LEOP, nominal and contingency passes.	Operations training quiz
11	*No separate lecture: the briefing opens the block.*	**Simulation 1 — LEOP and commissioning (6 h).** L12 briefing; Team A operates a full shift, Team B observes and documents; real-time anomaly injection; post-sim debrief.	Operators: sim log and anomaly tickets · Observers: critique notes
12	*No separate lecture: the briefing opens the block.*	**Simulation 2 — nominal operations and contingency (6 h).** L13 briefing; teams swap operator and observer roles; post-sim debrief.	Operators: sim log and anomaly tickets · Observers: critique notes
Final session	**L14 Debrief, synthesis and pathways (3 h).** Debrief led by the student Flight Directors; observer presentations; lessons-learned synthesis; Canadian space sector pathways; course evaluation; **final mission review** (cohort-internal).		Final mission review; observer critique report; **individual reflective portfolio due 1 week later**

Regulation on Bilingualism at the University of Ottawa

Every student has the right to require that a course be given in the language used to describe the course in the course calendar (Regulation on Bilingualism at the University of Ottawa). Except in programs and courses for which language is a requirement, all students have the right to produce their written work and

to answer examination questions in the official language of their choice, regardless of the course's language of instruction.

Notice of Collection of Personal Information for Recording of Lesson Capture

Following the *Freedom of Information and Protection of Privacy Act* in Ontario and University Policy 90, your personal information is collected under the authority of the *University of Ottawa Act, 1965*. Any Zoom sessions and simulation video recordings will be retained for purposes consistent with the fulfilment of the course learning activities and outcomes, and may include your name, appearance, image, voice, and messages. If you attend remotely and do not wish to be recorded, you may disable your audio and video. If you attend in person and do not wish to be recorded, please contact the instructor during the first week of class to arrange alternatives. The Cyberrange is intended to be used with cameras for team observation. The information collected following this notice will be retained for one year from the end of the semester.

Generative AI and Attribution

Generative AI use is permitted, and in some activities encouraged, subject to the AIG (Assisted by Generative AI) transparency framework introduced in Week 1.

AIG / GIA attribution: Peters (2023), Logos IA-EN, CC BY-NC-SA 4.0 — `https://mpeters.uqo.ca/en/logos-ia-en-peters-2023/`

You will be expected to attach the AIG badge (or its French form GIA) to any deliverable that used generative AI, along with a short notes section describing how it was used. Passing off AI-generated work as your own, without the attribution, is an academic-integrity violation. Refusing to use AI where it is permitted is fine; what is not acceptable is using it without saying so.